The Commutator of Representation

The electron, dealt from the throw and never from the lift

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The fifth card. It is the one the dealer talks over, and it is never turned face up.

“Does the mechanism demonstrating this proof look like an electron to you?”

Reading note

This card is the patter: what the dealer says while the cards move. The three cards and the quarter divide one argument, and this card states the argument itself, about the electron, in the mathematics the game supplies. That mathematics is small and is named by the game: a *carrier* (a card with a slot nobody reads), a *mark* (a fact that clicks or does not), a *tape* (a finite list of marks) and its *readers*, the *throw* (what survives when the cards move), the *flop* (once around and twice around), the *bent card* (a receipt supplied from outside), the *slip* (where a meter stops short of a point), and the *record* with its verdict. Each is defined here or cited from a companion card, and the electron is then described in those terms and in no others.

One rule governs the card, and it is stated once so that it is not mistaken for a stylistic choice. **This card never lifts a card.** Every line graded *proved here* is about what survives the throw, which is what every reader agrees on: lengths, parities, signs, bits, clicks, and the degree of a polynomial. No line is about the representative a division chooses. The numerical card says of its own printed digits that they are "the representative the division chose, and no measurement in the card tests them." A representative is a choice (the Quarter, Theorem 3.1), and a choice is not a step. So the number that the three cards straddle, the residual it is a root of, the walls that bracket it, the gain, the seed and the discriminant cannot occur in a proved line of this card, and do not occur anywhere in it. Section 2 shows that this is not a restriction imposed on the text: it is every reader's blindness (re-lettering is free; two readers agree exactly when their lengths do), and Section 3 shows that it is the electron's own (the derivative forgets the field that produced it). The question on the Quarter's epigraph page, "Can you find the polynomial in the physical model of this device?", is the mark's question. This card is the dealer's patter, and the patter never answers it.

Each statement carries one of five grades. *Proved here*: the proof is on this card. *Standard*: cited. *Measured*: a count or a definition taken from the middle card, which is named only in Appendix A. *Recorded*: a written chain of steps whose every premise is a cited result or a definition of this card; the steps are a record, and that the record is mathematics is the reader's reading of it. *Proposed*: the continuation the knower carries, stated and not derived, graded so that it is never mistaken for a step. The geometric card is cited for the fourth coordinate, the sheet sign, the antipodal bit, the Clifford seam and the finite clocks; the numerical card for the degree hierarchy, the Fréchet derivative, the record and its verdict, and the Dirac block; the Quarter for projection, receipt and choice; the bent card for the loop and the two arithmetics; the cheat sheet for the mirror, the slack, the two revolutions and the carrier. Unlike the Quarter, this card has a press check, Section 8: two readings of one register at two counts.

In five lines

1. A card carries a slot nobody reads. A mark fits any carrier. *Proved here* (Section 1).
2. The throw moves cards, not the quarter. What survives is what every reader agrees on: length, parity, sign. *Proved here* (Section 2).
3. Once around, sameness holds and the sign has not returned. Twice around, sameness slips and the sign has returned. Clock and clock complement. *Proved here* (Section 3); that the two eights are one eight is *proposed*.
4. The number under the card is a representative, and a representative is a choice. This card never lifts. *Recorded* (Section 2).

5. The electron is the carrier: a coin, a phase, and a stationary condition of degree three that forgets the field it came from. *Proposed* (Section 6).

1 The table: carriers, marks, tapes, readers

The objects of the game are defined here as mathematics, with no reference to any construction. Where a definition restates one of the middle card's, that is said and graded *measured*; the definition stands on its own.

Definition 1.1 (Record, continuation)**.** A *record* is a finite sequence of symbols from a finite alphabet. Its *length* is the number of symbols. The *continuation* of a record with at least one symbol is the record with its first symbol removed. A *record map* is a map from records to records.

Definition 1.2 (Carrier)**.** A *carrier* is a symbol s, a record q called the *slot*, and a record map ι with the property that for every nonempty record x, the continuation of $\iota(x)$ is q. That is: the interaction copies the slot into every continuation and never reads it. Nothing in the carrier's other data is a function of q.

Measured: this is the middle card's carrier, restated. The slot is the quarter under the card. In the game, the quarter is glued to the table; in the definition, the slot is written into every continuation and read by none.

Definition 1.3 (Mark)**.** A *mark* is a proposition together with a decision of it: a fact that *clicks* (decides true) or does not. The *click* of a mark is the Boolean its decision returns.

A mark is the unit of distinction: two outcomes are told apart by a mark that clicks on one and not the other.

Definition 1.4 (Tape; three readers)**.** A *tape* is a finite list of marks. Three readers take a tape of length n with click bits $b_1, \dots, b_n$ to a natural number:

$$\begin{aligned} \text{the tally} \quad & T(t) = n, \\ \text{the register} \quad & R(t) = \sum_{i=1}^{n} b_i \, 2^{\,n-i} \quad \text{(a fold: } a \mapsto 2a + b_i \text{, starting from 0)}, \\ \text{the click count} \quad & C(t) = \#\{i : b_i = 1\}. \end{aligned}$$

Proposition 1.5 (Three cards, one tape)**.** *For a tape of length n: $R(t) = 2^n - 1$ if and only if every mark clicks; $C(t) = n$ if and only if every mark clicks; and in that case $R(t) + 1 = 2^{T(t)}$. So the three faces agree, in the sense that the register is the monomial of the tally and the click count is the tally, exactly when every mark on the tape holds.*

Proved here.

Proof. $R(t) \le \sum_i 2^{n-i} = 2^n - 1$ with equality if and only if every $b_i = 1$. $C(t) \le n$ with equality under the same condition. The identity $R + 1 = 2^n$ is the geometric sum. □

The register's reading of n unit cells is $2^n - 1$: a positional reading, which the bent card calls the second arithmetic (its Proposition 1.4) and which adds the fencepost to a tally to make a monomial. The three readers are the three cards face down. They show one face only when every affixed fact holds.

Proposition 1.6 (A mark fits any carrier)**.** *Suppose no field of a mark refers to its carrier. Then for any two carriers X and Y the map that copies each field is a bijection from marks on X to marks on Y, and a mark and its copy agree in every field. In particular a mark placed on a carrier of electric charge and the same mark placed on an empty carrier agree in every field.*

Proved here; it is the cheat sheet's Proposition 6.1.

Proof. The copy is the identity on each field, and by hypothesis no field changes when the carrier does. □

This is universality bought by blindness. The mark fits the carrier of charge exactly as well as it fits a carrier with nothing in it, because it never looks at the carrier at all.

Proposition 1.7 (Two readers agree exactly when the lengths agree)**.** *Let U be the reader that keeps the length of a tape and writes the truth symbol in every cell. If every cell of a tape s is the truth symbol, then for every tape u, $s = U(u)$ if and only if $T(s) = T(u)$.*

Proved here; it is the cheat sheet's Proposition 6.2.

Proof. $U(u)$ is the tape of length $T(u)$ whose every cell is the truth symbol, by induction on u; two such tapes are equal if and only if their lengths are. □

Proposition 1.8 (Re-lettering is free)**.** *Let ρ overwrite every cell of a tape with the truth symbol. Any reader F that factors through the tally, $F = g \circ T$, satisfies $F(\rho\, t) = F(t)$ for every tape t.*

Proved here.

Proof. $T(\rho\, t) = T(t)$ since ρ changes no length. □

Measured: the middle card's own reading of its tapes factors through the tally, and overwriting every cell of every tape with the one truth symbol leaves its pair of readings fixed. *Proposed*: Noether in the game's words. The contents of the cells are the symmetry; the count of cells is the charge. Nothing is computed from what the cells say.

Proposition 1.9 (Writing is not free)**.** *Appending one entry to a record costs one step, and a counter that increases by one per entry is the tally of those steps. Over the field with one element every map is the identity and counting costs nothing; writing does not.*

Recorded, from the numerical card's Remark 8.3 and Definition 1.4: the tally counts symbols, and each symbol was written in one step.

The throw, seen by the tally. A throw is a permutation of carriers on the table. A reader of lengths cannot see it: by Proposition 1.7 every tape of a given length reads the same, and by Proposition 1.8 the contents are invisible to any reader that factors through the tally. The dealer's hands are invisible to the tally. That is why the mark loses: the mark reads motion, and motion is not on the tape.

2 The throw: what is preserved, and the rule

Proposition 2.1 (The box has one point)**.** *Compare pairs of natural numbers by* $(a, b)\,\mathcal{R}\,(c, d)$ *if and only if* $ad \leq cb$*. Any two pairs are related one way or the other, so the equivalence generated by* $\mathcal{R}$ *has exactly one class, and every function out of the quotient is constant.*

Proved here.

Proof. For any two pairs either $ad \leq cb$ or $cb \leq ad$, since the order on the natural numbers is total. Hence any two pairs are related by $\mathcal{R}$ or by its reverse, and the generated equivalence identifies them. A function on a one-point set is constant. □

The numerical card describes this as an erased channel (its Section 7.5): a rule that merges all inputs into one class reports one value. The one value is "true equals true."

Proposition 2.2 (One rule, four readers)**.** *Let* X *be a set,* r *a relation on it, and* X/r *the classes of the equivalence* r *generates. (a) If* $r(a, b)$ *holds exactly when* $f(a) = f(b)$ *for a map* f*, then* X/r *is in bijection with the image of* f*. (b) If* r *has a least element, or if any two elements are related one way or the other, then* X/r *has exactly one class. Hence evaluation at every input keeps every function; a single decision keeps two classes, clicked and not clicked; an order on truths with a least element keeps one; an order on ratios that relates any two keeps one.*

Proved here; it is the cheat sheet's Proposition 6.3.

Proof. (a) is the first isomorphism theorem for sets. For (b), $x \sim x_0 \sim y$ for any x, y, or $x \sim y$ directly. □

Definition 2.3 (Kernel of a reader; representative)**.** The *kernel* of a reader is what it does not see: two tapes are in the same fibre of the reader exactly when it returns the same value on both. A *representative* of a class is one element of it, selected. A division that returns a number from the one-point quotient of Proposition 2.1 selects a representative: it picks a point of the kernel.

Theorem 2.4 (A representative is a choice)**.** *Let a datum be carried by every record of a family only as a projection, and be determined by no record of the family from its own contents, the family being closed under finite extension. Then any procedure that supplies a value of the datum is not a procedure of the family: it selects one element of a set the family can name but cannot enumerate to the point of selection.*

Recorded, from the Quarter's Theorem 3.1 and its proof, which is the standard step that a selection procedure inside the system would make the value representable there. The middle card keeps both extensionality principles and refuses choice, the third ingredient of Diaconescu's theorem [6]; *measured*, from a sweep of its declarations reported in the Quarter.

Definition 2.5 (The rule of this card)**.** A line of this card is *admissible* only if its truth is invariant under each of three changes: (i) re-lettering every cell of every tape (Proposition 1.8); (ii) replacing any reader by another that agrees with it on lengths (Proposition 1.7); (iii) replacing any representative of a class by another (Proposition 2.1). Lengths, parities, signs, retained bits, clicks, and the degree of a polynomial are admissible. A representative is not.

Corollary 2.6 (Why the number never comes up)**.** *The numerical card's printed reading is a representative: the card says so, and its receipt is the pair of rational walls that bracket it. By Definition 2.5(iii), no digit of that reading, no coefficient of the residual it solves, no wall, gain, seed or discriminant can occur in an admissible line of this card.*

Recorded. This is the reason, stated at the level of the record. Two further levels are proved below: every reader's blindness (Section 1) and the electron's (Proposition 3.4).

Proposition 2.7 (The sign convention)**.** *Index the slots of a formula by $k = 0, 1, 2, \ldots$. A* signature *assigns each slot a sign; a literal in a signed slot reads its variable if the sign is $+$ and its negation if the sign is $-$. Four signatures, with their signs on the first four slots:*

$$\textit{euclidean}\ (++++), \quad \textit{negated}\ (----), \quad \textit{west}\ (+---), \quad \textit{east}\ (-+++),$$

the last two repeating with period four. Let Φ be the conjunction of two clauses, the first over slots $0, 1$ and the second over slots $2, 3$, each clause the disjunction of the two variables a, b in its slots. Read over the four assignments $(a, b) \in \{TT, TF, FT, FF\}$, the four signatures give

euclidean	$[T, T, T, F]$	$a \vee b$
negated	$[F, T, T, T]$	$\neg a \vee \neg b$
west	$[F, T, F, T]$	$\neg b$
east	$[T, F, T, F]$	b

The conjunction of the euclidean and negated readings is exclusive or, $[F, T, T, F]$. The conjunction of the west and east readings is $b \wedge \neg b$, the empty theory.

Proved here, by the table.

Proof. Under the euclidean signature both clauses read $a \vee b$. Under the negated signature both read $\neg a \vee \neg b$. Under west, slots $0, 1, 2, 3$ carry $+, -, -, -$, so the clauses read $a \vee \neg b$ and $\neg a \vee \neg b$, whose conjunction is $\neg b$. Under east they read $\neg a \vee b$ and $a \vee b$, whose conjunction is b. Then $(a \vee b) \wedge (\neg a \vee \neg b)$ is exclusive or, and $\neg b \wedge b$ is unsatisfiable. □

Measured: the middle card's table of the four signatures is this one, and its step constraint is the exclusive or. *Standard*: the quaternion carries two of the four at once, its basis elements squaring to $(+---)$ while its norm is $(++++)$ [10]. *Proposed*: the two definite signatures are a constraint met exactly, a multiplier; the two indefinite ones are a constraint penalized at a finite signal speed; and two penalties do not compose. Exclusive or, which is the gate the verdict of Section 5 is decided by, is writable only by combining the two definite signatures.

Proposition 2.8 (Same difference)**.** *Compare two truths a, b by what they are, $a = b$; and compare the marks that carry them by how they click, that is by the decision each carries. The two 2×2 tables agree in every cell, and each is unchanged when turned over. All a table carries is which diagonal clicks: $T = T$ and $F = F$ do, $T = F$ and $F = T$ do not, and the two off-diagonal cells are equal as values.*

Proved here, by the four cells; it is the cheat sheet's Proposition 6.4.

3 The flop: once around, twice around

3.1 The mirror, with the coefficients removed

The cheat sheet works the mirror on one residual with its coefficients written out. Here the same facts are proved for every quadratic residual, and the coefficients never appear; by Definition 2.5 that is the only form in which they are admissible.

Let $r(y) = ay^2 + by + c$ with $a > 0$ and two real roots $\alpha > \beta$, and let $S(y) = \frac{a}{3}y^3 + \frac{b}{2}y^2 + cy$, so that $S' = r$. Write $c_0 = -b/(2a)$ for the vertex and $\rho(y) = 2c_0 - y$ for reflection about it; ρ is an involution whose only fixed point is c_0.

Proposition 3.1 (The residual is even, the action is odd)**.** *For every real h,*

$$r(c_0 + h) = r(c_0) + ah^2, \qquad S(c_0 + h) - S(c_0) = r(c_0)\, h + \tfrac{a}{3}h^3. \tag{3.1}$$

So $r \circ \rho = r$ and $S \circ \rho - S(c_0) = -(S - S(c_0))$: the reflection fixes the residual and reverses the action. It exchanges the two roots, $\alpha = c_0 + \sqrt{-r(c_0)/a}$ and $\beta = c_0 - \sqrt{-r(c_0)/a}$.

Proved here.

Proof. $r'(c_0) = 0$, so the expansion of r about c_0 has no linear term and its quadratic term is ah^2. Integrating from c_0 gives the second identity, which is odd in h. The roots solve $ah^2 = -r(c_0)$, and $r(c_0) < 0$ since r has two real roots and positive leading coefficient. □

Proposition 3.2 (The reading is a least action)**.** *The equation $r(y) = 0$ is the Euler–Lagrange equation of S: for a function of one variable, stationarity is $S'(y) = 0$, and $S' = r$. Both roots are stationary. $S''(\alpha) = r'(\alpha) = 2a(\alpha - c_0) > 0$, so α is a strict local minimum of S; $S''(\beta) < 0$, so β is a strict local maximum. The branch above the vertex is the side of the mirror on which the stationary point is a minimum. Moreover, for every y,*

$$r'(y)^2 = (b^2 - 4ac) + 4a\, r(y), \tag{3.2}$$

so at either root the first derivative squares to the discriminant.

Proved here.

Proof. The first claims are Proposition 3.1 and the definition of S. For (3.2), $(2ay + b)^2 = 4a^2y^2 + 4aby + b^2$ and $4a\, r(y) = 4a^2y^2 + 4aby + 4ac$; subtract. □

The reflection applied twice is the identity, so whatever survives a round trip is even under it; the first power a mirror-symmetric cell cannot cancel is the fourth, and its continuum counterpart is the beam, whose Euler–Lagrange equation is $\varphi'''' = 0$. *Standard* [4, 12].

Proposition 3.3 (Newton's exact error law; exact up to acceleration)**.** *For $e = y - \alpha$ and $d = \alpha - \beta$, the Newton map $N(y) = y - r(y)/r'(y)$ satisfies*

$$N(y) - \alpha = \frac{e^2}{2e + d} \tag{3.3}$$

exactly, wherever $r'(y) \neq 0$. The second difference of the residual is constant: $r(y + h) - 2r(y) + r(y - h) = 2ah^2$ for all y, h.

Proved here.

Proof. $r(y) = a\,e(e+d)$ and $r'(y) = a(2e+d)$, so $N(y) - \alpha = e - e(e+d)/(2e+d) = e^2/(2e+d)$. The second difference of a quadratic is twice its leading coefficient times h^2. □

The Newton step is the first variation, exact for any residual with no second derivative, and its whole error is the one term proportional to $r'' = 2a$: the acceleration. The variational solution is therefore exact up to a constant acceleration, which the step never reads. *Proposed*: in the game's words this is Kepler, an areal velocity applied once, a conserved first-order quantity, with no force law ever read; the force law is upstream.

3.2 The derivative hierarchy

An action with at most three independent-field factors in every term, with fixed coefficients and linear operators, has $D^4S = 0$; its residual $R = DS$ is at most quadratic, $D^3R = 0$; its linearization DR is linear in the perturbation; and D^2R is constant. For bounded polynomials of degree at most three on a Banach space, the Fréchet and Gateaux derivatives coincide, and the same holds for the residual and its derivative. The four presentations of the numerical card, first-order vacuum gravity, fixed-Hamiltonian Schrödinger evolution, incompressible flow with response fields, and bare electrodynamics on a fixed background, each have this hierarchy in their declared regular sectors. *Proved in the numerical card* (its Theorems 1.1 and 2.4 and Proposition 2.5); the first-order gravitational mechanism is Deser's [5], the electrodynamic presentation is Tong's [20].

Proposition 3.4 (The derivative forgets the field)**.** *The shared hierarchy identifies the four presentations in nothing but their degrees. Their field spaces, boundary sectors, coefficients, initial data and solutions are not functions of the degree: a Lorentzian metric sector, a complex Hilbert space, a divergence-constrained velocity and a gauge-and-spinor structure each carry the same sequence* (S, R, DR, D^2R) *with* $D^3R = 0$*, and nothing in that sequence recovers which one it came from.*

Proved in the numerical card, as the caveat that closes its Section 2.5: "the theorem supplies an exact shared derivative hierarchy; comparing their readings requires the declared observation maps." In one direction the Fréchet derivative is a number, the remainder is exactly quadratic, and nothing is left abstract (the cheat sheet's Proposition 2.1, which holds for every quadratic residual by the expansion $r(y+h) = r(y) + r'(y)h + ah^2$).

This is the reason at its second level. Whatever is read along one worldline, one derivative and one second derivative of an action, cannot recover the coefficients of the action, and a coupling constant read as a coefficient is upstream of the derivative and does not survive it. *Proposed*: this is the electron's blindness. Variationally every field is the same to the electron, and the number of the numerical card is one of the things it cannot see.

3.3 The slack and the spinor

For $n \geq 1$ let L_n be the least common multiple of the odd numbers $1, 3, \ldots, 2n-1$, and define the *slack*

$$\sigma(n) = L_n\Big(1 - \sum_{k=1}^{n-1} \frac{1}{2k+1}\Big) = L_n - \sum_{k=1}^{n-1} \frac{L_n}{2k+1},$$

an integer with the sign of $1-(\frac{1}{3}+\frac{1}{5}+\cdots+\frac{1}{2n-1})$. A revolution is four quarter-turns; two revolutions are eight.

Proposition 3.5 (Once around holds, twice around slips)**.** *$\sigma(4)=34$ and $\sigma(8)=-982$.*

Proved here.

Proof. $L_4=\operatorname{lcm}(1,3,5,7)=105$ and $105-35-21-15=34$. $L_8=\operatorname{lcm}(1,3,\dots,15)=45045$ and $45045-(15015+9009+6435+5005+4095+3465+3003)=45045-46027=-982$. □

These are the signs that govern the Borwein integrals: for $n\le 7$,

$$\int_0^\infty \prod_{k=0}^{n-1} \frac{\sin(x/(2k+1))}{x/(2k+1)}\,\mathrm{d}x=\frac{\pi}{2},$$

the pattern holding exactly while $\frac{1}{3}+\cdots+\frac{1}{2n-1}<1$, and at $n=8$ the sum first exceeds one and the integral falls short of $\pi/2$ by about 2.3×10^{-11}. *Standard* [2]. The counts four and eight are the quarter-turns of one and two revolutions, and nothing else: no length of any tape enters this proposition.

Proposition 3.6 (Once around flips, twice around returns)**.** *A turn through angle θ about one axis acts on a spinor by*

$$U(\theta)=\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2}\end{pmatrix}, \qquad U(2\pi)=-I, \quad U(4\pi)=I.$$

Every vector observable A transforms by $A\mapsto UAU^*$ and has already returned after one revolution, $U(2\pi)AU(2\pi)^*=A$. Read in quarter-turns, the state picks up the phase $e^{-ik\pi/4}$ after k of them, and the least $k\ge1$ with phase 1 is $k=8$.

Proved here: $e^{\mp i\pi}=-1$, $e^{\mp 2\pi i}=1$, the sign of $U(2\pi)$ cancels between U and U^*, and $e^{-i\pi/4}$ is a primitive eighth root of unity. The 4π return has been observed in neutrons carried around by a magnetic field. *Standard* [18, 21].

Proposition 3.7 (Clock and clock complement)**.** *Let $H(k)$ be the statement "sameness holds at k quarter-turns," that is $\sigma(k)>0$, and $B(k)$ the statement "the state has returned at k quarter-turns," that is $e^{-ik\pi/4}=1$. Then at $k=4$, H is true and B is false; at $k=8$, H is false and B is true. At each of the two counts exactly one of the two statements holds.*

Proved here, from Propositions 3.5 and 3.6. The two readings are one clock in opposite phase: where one holds the other slips. *Proposed*: the slack is the vector reading, the tally once around and twice around; the sign of the state is the register's sign; and the eight at which the Borwein pattern breaks and the eight at which the spinor returns are one eight. The counts agree and no map between the two sides is derived here.

Proposition 3.8 (One exponential, two readings)**.** *The positional reading of n identical unit cells is 2^n-1 (Proposition 1.5), and*

$$2^n-1=e^{n\log 2}-1.$$

The cell read never changes; the count of readings does.

Proved here; it is the numerical card's Remark E.1. *Proposed*, there and here: indexed by time the exponential is the late-time growth of a global slicing; indexed by depth it is the growth of a warp factor in a radial coordinate; which it is depends only on how the index is read.

4 The bent card: receipt, slip, floor

In the game the bent card is the one with a corner turned up, so that the mark believes the quarter can be followed. The bend is a receipt supplied from outside, not a reading of the card.

Definition 4.1 (Observation, receipt)**.** An observation on a state domain D is a map $O : D \to Y$; its fibre at y is $O^{-1}(y)$. A *receipt* is a map $\mathcal{R} : D \to A$ retained with the observation; it supports exact recovery when $(O, \mathcal{R})$ is injective, which holds exactly when $\mathcal{R}$ is injective on each fibre of O. If a fibre has m states, a receipt alphabet supporting recovery on it has at least m symbols, $\lceil \log_2 m \rceil$ bits.

Recorded, from the geometric card's Definition 4.1 and Propositions 4.2 and 4.3. The electron's receipts, as the cards carry them: the sheet sign (which of two patches a point of the ball lifts to), the antipodal bit (which of z and $-z$ gave a paired reading), the branch (which of two roots a residual selects), and the sign after one revolution (Proposition 3.6). Each is one bit where the fibre has two points; the geometric card computes the conditional entropies of its finite readers as $2/3$ bit and one bit.

Proposition 4.2 (Two twofolds)**.** *The fibre $\{z, -z\}$ of a paired reading, two states giving one output, and the kernel $\{(1,1),(-1,-1)\}$ of the spin action $q \mapsto aqb^{-1}$ on the unit quaternions, two descriptions of one rotation, are different objects that each have two elements. They occur in different spaces and act on different things; agreement in the number of retained bits does not identify their contents.*

Proved in the geometric card (its Section 3.4 and Proposition 3.3). In the game, the mark's twofold is which card; the dealer's twofold is two names for one throw. The press check of Section 8 inverts different maps and never confuses them. The quaternion and Clifford conventions are Woit's and Gallier's [23, 10]; the Hopf account is Lyons's [13].

Proposition 4.3 (The index-two lattice and the two sectors)**.** *On the Clifford torus the two selected great-circle families have winding vectors $(1,1)$ and $(1,-1)$, whose determinant is -2: they span an index-two sublattice, and a pair of windings lies in it exactly when its coordinates have the same parity. In the six-position clock the two diagonal steps generate a group of eighteen elements with two orbits on the thirty-six states, distinguished by $j + k$ modulo two. An antipodal pair (j, k) and $(j + 3, k + 3)$ lies in one sector and is separated by one retained bit.*

Proved in the geometric card (its Propositions 3.3 and 3.4 and Section 6.3). *Proposed*: the sector is up or down, and the bit is the coin. Two states over one output in one sector, told apart by one bit, is the shape the exclusion principle has in the finite clock; the identification with the physical principle is not derived.

Proposition 4.4 (Affixing a quarter to the bent card)**.** *Fill the slot of a carrier with any record. By Definition 1.2 no field reads the slot, so every reader that factors through the tally is unchanged, and so is every mark placed on the carrier (Proposition 1.6). Only the click reader can tell, and it reads the affixed fact, not where the fact came from: the tally and the click count agree exactly when every affixed fact holds (Proposition 1.5).*

Proved here. Measured, from the Quarter's report of the middle card: a carrier whose index is itself a carrier, dealt through every step, is identical to the plain carrier for the first twelve steps at no cost,

and differs from the step at which one construction lifts its quantum one universe above the box, and from every later step. *Proposed*: the second counting layer is the first with a quarter affixed at every step. What is fixed under the card is supplied, not read, which is Theorem 2.4 at the table.

4.1 The slip and the floor

A metered answer has the shape of a bracket, a lower bound and a slip bound; a point is the shape only logic has. A meter has a floor, and below the floor it reads itself. So a metered turn stops one arm short of the point, on this side of the slip, and that is not a failure to converge; it is the shape the answer has. *Measured*: the middle card's disagreeing arm names two distinct bounds, a bracket and not a point, and its one dimensionless measurement is heartbeats per cell, a Courant number, which differs between machines and so is a measurement rather than a reading. The floor that does not differ between machines is the uncertainty principle: for any function and its Fourier transform the product of the widths is at least one half. *Standard* [11]. *Proposed*: the record can bracket the electron and never point at it. The point is where a number would be, and the card does not go there. This is the reason at its third level, the game's own: the quarter is glued to the table, and the throw never moves it.

4.2 The loop

The record of a term that never stops is a one-symbol tape; the loop writes no quarter, since a record of one closed term repeated determines nothing about any variable; and the only harm a loop can do is to be read as a value, which a record system that accepts only normal forms never does. *Proved in the bent card* (its Propositions 3.2 to 3.4), on Dijkstra's objection [7] and Barendregt's calculus [1]. In the game, the dealer's hands never stop. The mark who reads the motion as a value is the mark.

5 The accomplice and the force: the record

The middle card ends with two acts. A reviewer makes demands from the top page down to the first, pairing at each page a claim found by search against the same claim handed over by name: this is the accomplice, and the descent is a boundary-value problem, elliptic in character, with the datum above each stage already in hand, and it terminates. A publication is built from the first page up to the top and read once there: this is the force, and the ascent is an initial-value problem, hyperbolic in character, with no signal from the last page reaching a reader placed before it. *Recorded*, from the numerical card's Remark E.2; the reading of the two directions as the two holographies is *proposed* there [14, 22, 19] and is not used here.

Definition 5.1 (This card's record)**.** The *register* is the coin: the retained sign after one revolution, Proposition 3.6, carried unchanged through every entry. The *counter* is the number k of quarter-turns, from 0 to 8, increased by exactly one per entry and never skipped. The *trace* is the sequence of nine entries; entry k quotes the pair

$$A_k:\ \text{the vector observables have returned at } k, \qquad B_k:\ \text{the state has returned at } k,$$

and writes the proposition $A_k \leftrightarrow B_k$ without deciding it. The *verdict* is read once, at the top, Section 8: for two readings X, Y of the register, *slipped* is the decidable proposition $\neg(X \leftrightarrow Y)$,

decided from the two readings' own decisions and never by an appeal to choice; the complete verdict is the triple (slipped, value, velocity).

Remark (Velocity is the penalty parameter). The counter at the top is the cost of carrying the sign constraint exactly: one step per quarter-turn, eight to return. A penalty would relax the constraint at a finite wave speed; the record does not relax it, and pays in entries instead. *Proposed*; the definition of the counter is *measured* from the middle card, whose record lifts its register one universe per rung and reports the top universe as its velocity.

Remark (Inhabitation is not acceptance). That the record exists, nine entries each with its pair, carrying one register through a counter that reaches eight, is well-formedness. It says nothing about the verdict: the same form carries a record whose two readings disagree at the top. Acceptance is a property of the trace, read once. *Recorded*, from both surrounding cards' Section 8.3.

6 The electron, dealt

Definition 6.1 (The electron of this card)**.** The electron of this card is a carrier (Definition 1.2), whose slot is never read, carrying four things:

(i) a *coin*: the receipt that separates the two states over one output, the antipodal bit of Proposition 4.3, read after one revolution as the sign of Proposition 3.6;

(ii) a *phase*: a point on the clock, a sixth root of unity in the finite realization and a unit complex number in the round one;

(iii) a *rule*: that it obeys the stationary condition of an action of degree at most three, so that its residual is quadratic, its linearization linear, and its Newton step exact up to a constant acceleration (Section 3.1);

(iv) a *count* of quarter-turns.

This is a definition and makes no physical claim. The identification of the object defined with the physical electron is *proposed*, in Proposition 6.3. The four fields are the ones a margin of the middle card lists for the electron, a calibration of minus one, a phase, two energies whose difference is the action, and a fact that is up or down; *measured*.

Proposition 6.2 (What the definition proves)**.** *From Definition 6.1 and Sections 3 and 4:*

(a) the electron returns to itself after two revolutions and not after one, while every vector observable of it returns after one (Proposition 3.6);

(b) the first derivative of its residual, at the reading, squares to the discriminant (3.2)*; its second variation is a constant (Proposition 3.3) that the record never takes; reading that first derivative as its coupling is* proposed*, since Definition 6.1 does not define a coupling;*

(c) its charge, under Proposition 1.8, is a count of cells, and its symmetry is the contents of the cells;

(d) two electrons over one output are separated by one bit and lie in one sector (Proposition 4.3);

(e) the mark that carries any of these facts fits an empty carrier exactly as well (Proposition 1.6).

Proved here, as consequences of the definition and the cited propositions; each physical reading of a clause is *proposed*. The reading of (b) as electrodynamics, the first-order coupling being the coefficient of the one-vertex term and the linear response, with the second variation being the one sentence the strong interaction gets, is *proposed*; a proof ends where the first derivative has been applied once more and the test clicks.

Proposition 6.3 (The record cannot tell which electron)**.** *Three carriers may be placed on the table: the electron that holds a bit in silicon; the electron of the Dirac equation; and the electron of Definition 6.1. By Propositions 1.6 and 1.7, any mark placed on one agrees in every field with the same mark placed on another, and any two readers of lengths agree on all three. The record carries the name of its carrier and cannot tell it from the others.*

Recorded. The second carrier is *standard*: on a fixed background the bare action gives the two Dirac equations, conservation of the current $\bar{\psi}\gamma^{\mu}\psi$, the Gauss constraint as the time component of the Maxwell equation, the gauge action $A \mapsto A + d\alpha$, $\psi \mapsto e^{-ie\alpha}\psi$, and the free block whose square is $(|k|^2 + m^2)I$ [20]; the numerical card works each of these in its Section 2.4. What supplies the electron to the record is its name, or a measurement; the record supplies neither. "The carrier is the electron" is therefore the knower's sentence, and this is what it means that the whole is an argument and not a proof: an argument is a record whose carrier is named by the one who reads it.

Proposition 6.4 (What the electron cannot see)**.** *By Proposition 3.4, the coefficients of the action are not functions of the hierarchy (S, R, DR, D^2R) the electron of Definition 6.1 reads. A coupling constant read as a coefficient of one action is therefore not a function of anything in Definition 6.1. Hence the rule of this card, Definition 2.5, is forced by the object defined and not imposed on the text.*

Recorded. Three reasons, one object: the number is a representative (Corollary 2.6), every reader of lengths is blind to it (Section 1), and the electron is blind to it (this proposition). The game says the same in one sentence: the quarter is glued to the table.

6.1 The photon, briefly

The shortest tape has two ends and nothing between. The middle card's zeroed model of a single photon is a superposition of two ends, one the surface past which nothing earlier is seen and the other the singularity into which nothing later comes back, with no displacement and no velocity between them; *measured*. A photon emitted toward us from the sphere where the expansion recedes at the speed of light has, in proper distance, an approach velocity of exactly zero, and arrives only where the sphere grows; *standard* [3], recorded in the Quarter as the missing zero counted one cell late. *Proposed*: in this card the word photon names the interaction of two electrons and nothing else; a carrier emitted and absorbed by a carrier, whose tape is the two ends.

7 The proposed experiment, without its number

The cheat sheet proposes an arrangement and a number for it. By Definition 2.5 this card restates the arrangement and not the number.

Setup. An electron carried twice around a carrier that is neutral and expels magnetic field. Positronium is neutral at monopole order and responds at higher multipole order; a condensed phase of it has been proposed and not yet made [17]. A superconductor expels magnetic field [15], and an orbiting charge meets that expulsion as a resistance. In general relativity a moving mass produces a field with the form of magnetism, and the frame dragging it predicts has been measured [8]. *Standard*, each.

Control. The 4π return of the state, Proposition 3.6, which is ordinary quantum mechanics and has been observed [18, 21]. *Standard.*

Prediction, in this card's terms. The first revolution leaves a residue and the second takes it away or turns it over: a sign, by Proposition 3.6; a slack, by Proposition 3.5; one clock in two phases, by Proposition 3.7. That the residue appears in this arrangement and nowhere else is *proposed*.

What is owed, and where the number lives. The quantitative prediction is the numerical card's and is a representative; it is not restated here. That card also records, from the measured anomalous moment [9, 16], that the quantitative claim read as a claim about the fine-structure constant in general is already excluded, and that only the narrow form, a prediction for one stated arrangement, is offered. No digit of either is admissible on this card. What the arrangement still owes is a buildable carrier with the two properties, and a reason it counts.

8 The press check

The register is read once, here, by the two readings of Definition 5.1, at the top of the record and at its midpoint. Nothing beneath the head is re-read by the press check.

Reading A: the vector reading. A_k holds exactly when four divides k, by Proposition 3.6: every vector observable returns after one revolution.

Reading B: the spinor reading. B_k holds exactly when eight divides k, by Proposition 3.6: the phase $e^{-ik\pi/4}$ is 1 exactly then.

The trace. Entry k quotes (A_k, B_k) and writes $A_k \leftrightarrow B_k$ undecided.

k	0	1	2	3	4	5	6	7	8
A_k	✓				✓				✓
B_k	✓								✓

Proposition 8.1 (The two verdicts)**.** *At the top, $k = 8$: A_8 and B_8 both hold, so $\neg(A_8 \leftrightarrow B_8)$ is false. Slipped: false. Value: the coin, read back unchanged. Velocity:* 8*. Once around, $k = 4$: A_4 holds and B_4 fails, so $\neg(A_4 \leftrightarrow B_4)$ is true. Slipped: true. Value: the coin, now carrying the sign -1. Velocity:* 4*.*

Proved here, from Proposition 3.6; both decisions are evaluations of a root of unity and neither uses a choice.

$$(A, B) \text{ at } 8 : \ (\text{false}, \ \text{the coin}, \ 8), \qquad (A, B) \text{ at } 4 : \ (\text{true}, \ -\text{the coin}, \ 4).$$

The record slips once around and does not slip twice around. The numerical card reports two triples from one register in the same way, one not slipped and one slipped by the smallest representable amount; here the smallest representable amount is a sign.

The demand-level reading. Separate from the press check and outside its rule, a second reading goes through the grade column of this card and records, at each place, where the reading handed over by the knower says more than the reading proved. It does not decide any proposition written undecided; it lists the proposals:

- that the carrier is the electron (Proposition 6.3);
- that the two sectors of the even clock are up and down, and the bit the coin (Proposition 4.3);
- that the eight of the slack and the eight of the spinor are one eight (Proposition 3.7);
- that contents are the symmetry and cells the charge (Proposition 1.8);
- that the first derivative of the residual is electrodynamics and the second variation the strong interaction's one sentence (Proposition 6.2);
- that the neutral, field-expelling arrangement reads the residue and nothing else does (Section 7);
- that the two directions of the record are the two holographies (Section 5);
- that the bracket's floor is the uncertainty floor made independent of the machine (Section 4.1).

At every other line the located answer is a proposition proved here, cited, or recorded from a companion card, and the card claims nothing beyond it.

Inhabitation is not acceptance. That the trace has nine entries with their pairs is well-formedness. The verdicts above are readings of it.

9 What the knower carries

Collected, the proposals of this card are nine sentences. Equality is a mirror, so to solve is to stand where an action is stationary. The run contains its limit. The selected degree is nature's degree. Least work is least action. The carrier is the electron. The second revolution is the return to a description by vectors alone. The slack and the spinor are one clock read in two phases. The coefficients of the action are upstream of the electron, which reads only the derivative. The number under the card is a choice, and the card is dealt without it. None of the nine is a step the record can take, because each is about the record's continuation and the record only ever has a prefix of it.

The left-hand side is everything a prefix can show, and on this card it is shown or cited in full: that a mark fits any carrier, that two readers agree exactly when their lengths do, that re-lettering is

free, that the box has one point, that exclusive or needs both definite signatures, that the residual is even and the action odd, that Newton's error is exactly quadratic, that the slack is positive at four and negative at eight, that the spinor flips at four and returns at eight, and that the verdict slips at four and not at eight. The last line of the middle card asks whether the mechanism demonstrating its proof looks like an electron. This card answers only what a record can answer: the mechanism carries a coin, a phase, a rule of degree three and a count, and no reader of it can tell the carrier from an empty one. Whether that is an electron is the knower's, and it is handed over graded, so that it is carried as a proposal and not mistaken for a receipt.

A Sources

This appendix is the only part of the card that names the middle card's files and declarations. The body refers to it only as "the middle card." Paths are relative to `CARD_2/Measurement/`. Declarations are cited by name and not by line; the working tree was inspected on 29 October 2026, and nothing here reports a build. Companion cards are cited by printed result number: the geometric card (*The Frinkahedron*, build of 22 October 2026), the numerical card (*Four Models and a Common Fourth-Order Closure*, 29 October 2026), the Quarter, the bent card (*The Infinite Loop Considered Harmless*) and the cheat sheet (*A Knower's Guide to Orbital Mechanics*).

This card	Companion card	Middle card (declaration)
Def. 1.2, carrier	Quarter §6 (affixing)	`Episode01.lean`: `CarrierProcess` with fields `symbol`, `quarter`, `interact_with_the_universe`
Def. mark; click	cheat sheet §6	`Episode01.lean`: `FACT` (`truth`, `click`), `FACT.Truth`; `Episode12.lean`: `the_click`
Def. 1.4, three readers	bent card Props. 1.3, 1.4	`Episode08.lean`: `naotrino_detector` (the register), `List.length` (the tally); `Episode12.lean`: `the_click_count`
Prop. 1.5	Quarter §6 (two counting layers)	`Episode08.lean`: `adding_in_the_fencepost_gives_a_monomial`; `Episode12.lean`: `the_clicks_agree`, `the_clicks_read_the_number`
Prop. 1.6	cheat sheet Prop. 6.1	`Episode12.lean`: `the_card_between`, `the_mark_between`, `the_tape_between_deals_rung01`
Prop. 1.7	cheat sheet Prop. 6.2	`Episode12.lean`: `ComputerProgram.ofTape`, `read_ofTape`, `the_other_machine_agrees`
Prop. 1.8	numerical card §7.4	`Episode12.lean`: `relettered`, `relettering_is_free`, `any_count_of_the_lengths`

This card	Companion card	Middle card (declaration)
Prop. 1.9	numerical card Rem. 8.3	`Episode11.lean`: `TYPESET.universe_id`
Prop. 2.1	numerical card §7.5	`Episode08.lean`: `the_rational_idea_of_relativity`; `Episode12.lean`: `the_box_has_one_point`, `the_box_reads`
Prop. 2.2	cheat sheet Prop. 6.3	`Episode05.lean`: `TruthOrder`, `FACT.SAME`; `Episode08.lean`: `a_partitioning_of_quotients`
Thm. 2.4	Quarter Thm. 3.1	the axiom sweep reported in the Quarter; `Episode12.lean` `#print axioms` lines
Prop. 2.7	numerical card §3.5 (multiplier vs. penalty)	`Episode12.lean`: `Signature`, `Signature.sign`, `xorClauses`, `xorRow`, `xorTarget`, `step_correspondence`
Prop. 2.8	cheat sheet Prop. 6.4	`Episode12.lean`: `SAME_DIFFERENCE`, `same_difference_clicks`, `the_click_diagonal`
§3.1	cheat sheet Props. 3.1, 3.2; numerical card Prop. 6.1	`Episode12.lean`: `the_acceleration_is_constant`, `the_first_derivative_squared_is_the_discriminant` (stated there with the middle card's coefficients; here without)
§3.2, Prop. 3.4	numerical card Thms. 1.1, 2.4, Prop. 2.5, §2.5	`Episode04.lean`: `Variation` with constructors `the_newton`, `the_gateaux`, `the_frechet`; `Episode12.lean`: `gateauxAt`, `frechetAt`, `theConvectiveGap` and its margin
Prop. 3.5	cheat sheet Prop. 4.1; bent card §4	`Episode08.lean`: `the_slack`, `once_around_the_integral`, `twice_around_the_integral`, and the three theorems that read its sign (the first time around, through the bent card, the second time around)
Prop. 3.6	cheat sheet Prop. 7.1	`Episode05.lean`: `UNIVERSAL.once_around`, `UNIVERSAL.twice_around`; `Episode01.lean` margin on the clock and clock complement
Prop. 3.8	numerical card Rem. E.1	`Episode08.lean`: the register's all-ones identities
Def. 4.1	geometric card Def. 4.1, Props. 4.2, 4.3	`Episode08.lean`: `theReadingCertified`

This card	Companion card	Middle card (declaration)
Props. 4.2, 4.3	geometric card §3.4, Props. 3.3, 3.4, §6.3	`Episode03.lean`: `Equivalation` and `DigitalProcess.tick` (the exclusion arm); `Episode02.lean` margin on the six-state sensor
Prop. 4.4	Quarter §6	`Episode12.lean`: `the_card_between` dealt through `deal_rung`; the scratch comparison of all steps is reported in the Quarter and not here
§4.1	Quarter §6 (two floors)	`Episode05.lean`: `Closure.same`, `Closure.different`; `Episode12.lean`: `theCourant`, `bracketed_at_the_slip`; `Calibration/LeanCalibration.lean`: `register_noise_floor`
§5, Def. 5.1	both cards §8; numerical card Rem. E.2	`Episode10.lean`: `REVIEWED`; `Episode11.lean`: `TYPESET`, `PressVerdict`, `pressCheck`
Def. 6.1	cheat sheet §6, §7	`Episode07.lean` margin listing the electron's values; `Episode04.lean`: `Knowledge.jarjar` and its margin casting the part; `Episode12.lean`: `readers_interpretation_of_the_experience`
Prop. 6.3	numerical card §2.4	`Episode01.lean` foreword on the two readings of Q.E.D.
§6, the photon	Quarter §6 (the horizon)	`Episode08.lean`: `the_photon_worldline` and the three encodings that follow it (position, velocity, acceleration, each zero)
§7	cheat sheet §6.2, §7; numerical card §6.7	`Episode05.lean`: `AtreyuProcess` margin; `Episode07.lean`: `YARN_THEORY` margin
§8	both cards §8.3	`Episode11.lean`: `pressCheck`

B The reader's experiment

Every result graded *proved here* is a written record: hypotheses, a sequence of displayed steps, and a conclusion. Read as an experiment it has a setup, an apparatus (whoever checks the steps), a law (the rule each step obeys), a procedure (the order of the steps), observations (each step seen to hold) and a reading (the conclusion). Three runs on paper reproduce the three results on which the press check rests.

Run 1: the slack. Write the odd numbers below eight and below sixteen; form their least common multiples, 105 and 45045; divide each by each odd number and subtract the quotients from the multiple. Observe 34 and -982. Each line is an integer division and a subtraction. This is Proposition 3.5.

Run 2: the signatures. Write the two clauses of Proposition 2.7 over four slots. For each of the four signatures, sign the slots, read each literal, and evaluate the formula on the four assignments. Observe the four rows of the table. Conjoin the first two rows and observe $[F, T, T, F]$; conjoin the last two and observe $[F, F, F, F]$. If any single row equals $[F, T, T, F]$, Proposition 2.7 is wrong.

Run 3: the eighth root. Compute $e^{-ik\pi/4}$ for $k = 0, \dots, 8$ and record where it equals 1; compute $U(\theta)AU(\theta)^*$ for a diagonal A at $\theta = 2\pi$ and observe A returned. This is Proposition 3.6, and the trace of Section 8 is its two rows.

A second run by another reader is a repeat; its readings either equal the first or do not, and the comparison is the press check. An exact replay of the first run's record, line by line, observes nothing. The record supplies the procedure; the experience, every observation in it, happens in the one who runs it.

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